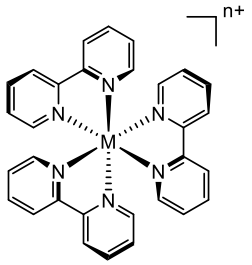
Inorganic Cumulative Exam
11-10-23
"Phun with Photons"

Instructions: Answer any five of the following ten questions. Only your first five answers will be graded. Each question is worth 10 points.

Molecular Inorganic Chemistry

1. Below is the generic structure for a metal tris 2,2'-bipyridine complex, $M(bipy)_3$. What is the point group of an $M(bipy)_3$ complex?



2. Consider the compound $Fe(bipy)_3^{n+}$. $Fe(bipy)_3^{n+}$ is commonly found in the 3+ and 2+ charge states. Assuming an octahedral ligand field, provide a d orbital splitting diagram / frontier orbital diagram for the high and low spin electron configurations of $Fe(bipy)_3^{2+}$ and $Fe(bipy)_3^{3+}$.

3. Which electron configuration (high spin or low spin) of $Fe(bipy)_3^{3+}$ do you expect to be the best oxidant? Explain your answer.

4. The UV/vis spectrum of $Fe(bipy)_3^{2+}$ contains a strong charge transfer band that is often described as a metal to ligand charge transfer (MLCT) band. Describe the excited state of $Fe(bipy)_3^{2+}$. What orbital does the electron start in before excitation? Where does the excited electron go?

5. Another common charge transfer band found in the UV/vis spectra of transition metal complexes is a ligand to metal charge transfer (LMCT) band. Propose a compound that you would expect to have an LMCT band in the UV/vis absorption spectrum. Why do you expect this compound to possess an LMCT band?

Solid State and Materials Chemistry

6. Describe the process of exciton formation in a photovoltaic device. What is required to form an exciton?

7. How can you convert a photovoltaic device, used to generate electricity, into a photodiode or a photodetector? Why will your proposed method be effective?

8. n-type / n-doped silicon was used in the construction of the first solar cell in 1954. Currently solar cells are constructed using p-type / p-doped silicon. Briefly describe "dopants" in semiconductors and give examples of dopants used to make p-type and n-type silicon.

9. Crystaline p-type / p-doped silicon solar cells generally exhibit a degradation of conversion efficiency during the first hours of exposure to the sunlight. This degradation of conversion efficiency is especially drastic in silicon containing relatively high levels of oxygen impurities. Propose a possible chemical reaction or process that occurs during light induced degradation of p-type / p-doped silicon.

10. How does a Schottky diode differ from a typical p-n junction semiconducting diode? How does the forward voltage drop of a Schottky diode compare to a p-n junction semiconducting diode?